

HR Analytics Dashboard Documentation

Cover Page

Project Title

HR Attrition & Retention Dashboard

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Tools Used

- Power BI
- Power Query
- DAX
- Python (Google Colab)

Dataset

Atlas Labs HR Dataset

1. Project Overview

Background

Employee attrition is a major challenge for organizations because it affects productivity, increases recruitment costs, and impacts employee morale. This project analyzes workforce data to identify patterns associated with employee turnover and provide recommendations that support better retention strategies.

The dashboard follows a business storytelling approach by answering four key questions:

- What is happening?
- Where is it happening?
- Why is it happening?
- What should HR do?

2. Objectives

The project aims to:

- Analyze employee attrition across the organization.
 - Identify departments with the highest turnover.
 - Examine the relationship between overtime, salary, experience, and attrition.
 - Perform correlation analysis to identify major attrition drivers.
 - Provide actionable HR recommendations.
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3. Dataset

Source

Atlas Labs HR Dataset

Tables used:

- Employee
- Education Level
- Performance Rating
- Satisfaction Level
- Rating Level

Total employees:

1,470

4. Data Preparation

The following steps were performed:

- Imported all tables into Power BI.
- Verified data types.
- Checked for duplicate Employee IDs.

- Checked for missing values.
 - Created relationships between tables.
 - Created DAX measures for KPIs.
 - Used Python to perform correlation analysis.
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5. KPIs

The dashboard includes:

KPI	Value
Total Employees	1,470
Attrition Rate	16.1%
Retention Rate	83.9%
Average Salary	\$112,956
Average Experience	4.6 Years

6. Dashboard Components

Describe each visual briefly:

KPI Cards

Provide a quick overview of workforce health.

Attrition Distribution

Shows the proportion of employees who left versus those retained.

Attrition Rate by Department

Compares attrition across departments.

Overtime vs Attrition

Illustrates the relationship between overtime and employee turnover.

Salary vs Years at Company

Shows salary trends across employee tenure.

Top Attrition Risk Drivers

Displays the strongest factors associated with employee attrition based on correlation analysis.

HR Action Plan

Provides business recommendations based on the findings.

7. Key Findings

- The organization has an attrition rate of **16.1%**.
 - A total of **237 employees** have left the company.
 - Promotion opportunities show the strongest relationship with employee attrition.
 - Employees with fewer years at the company are more likely to leave.
 - Overtime appears to be associated with increased attrition.
 - Salary has a relatively weak relationship with attrition.
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8. Recommendations

Based on the analysis:

- Improve promotion and career progression opportunities.
 - Strengthen manager–employee relationships.
 - Focus retention initiatives on early-career employees.
 - Reduce excessive overtime through better workforce planning.
 - Continue investing in employee development programs.
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9. Skills Demonstrated

- HR Analytics
 - Power BI Dashboard Development
 - Power Query
 - DAX
 - Data Cleaning
 - Data Visualization
 - Correlation Analysis (Python)
 - Business Storytelling
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10. Conclusion

This project demonstrates how workforce data can be transformed into actionable business insights. By combining Power BI with Python-based correlation analysis, the dashboard helps HR teams understand attrition drivers and prioritize retention strategies.